KAIZEN MATHS

Lesson starter

Hypothesis Testing and Inference

Proportions & Mixed Tests

Name

S1

In a sample of 120, 20 are defective calculators. Test at the 10% level whether the proportion of calculators that are defective has decreased from 0.25. Use a normal approximation for the sample proportion.

S2

In a sample of 100, 69 are students choosing the new revision platform. Test at the 1% level whether the proportion of students who choose the new revision platform is different from 0.6. Use a normal approximation for the sample proportion.

S3

Choose a suitable hypothesis test and carry it out. A sample of 27 gives 6 seeds that germinate. The original claim is that the germination probability is 0.45. Test at the 10% significance level whether the proportion has decreased.

S4

In a sample of 200, 56 are seeds that germinate. Test at the 1% level whether the proportion of seeds that germinate has increased from 0.25. Use a normal approximation for the sample proportion.

Show your working

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KAIZEN MATHS

Exit ticket

Hypothesis Testing and Inference

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Name _____

EA1

In a sample of 80, 44 are defective calculators. Test at the 5% level whether the proportion of calculators that are defective is different from 0.6. Use a normal approximation for the sample proportion.

EA2

Choose a suitable hypothesis test and carry it out. A sample of size 81 has mean 74.67 g. The population standard deviation is known to be 20 g. Test at the 10% level whether the population mean packet mass has decreased from 80 g.

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Exit ticket

Hypothesis Testing and Inference

Proportions & Mixed Tests

Name _____

EB1

In a sample of 120, 31 are students choosing the new revision platform. Test at the 10% level whether the proportion of students who choose the new revision platform has decreased from 0.3. Use a normal approximation for the sample proportion.

KAIZEN MATHS

Exit ticket

Hypothesis Testing and Inference

Proportions & Mixed Tests

Name _____

EA1

In a sample of 80, 44 are defective calculators. Test at the 5% level whether the proportion of calculators that are defective is different from 0.6. Use a normal approximation for the sample proportion.

EA2

Choose a suitable hypothesis test and carry it out. A sample of size 81 has mean 74.67 g. The population standard deviation is known to be 20 g. Test at the 10% level whether the population mean packet mass has decreased from 80 g.

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KAIZEN MATHS

Exit ticket

Hypothesis Testing and Inference

Proportions & Mixed Tests

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EB1

In a sample of 120, 31 are students choosing the new revision platform. Test at the 10% level whether the proportion of students who choose the new revision platform has decreased from 0.3. Use a normal approximation for the sample proportion.

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Teacher answers & working

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S1

In a sample of 120, 20 are defective calculators. Test at the 10% level whether the proportion of calculators that are defective has decreased from 0.25. Use a normal approximation for the sample proportion.

Answer:

reject H_0

Working:

Let p be the proportion of calculators that are defective.

H_0: $p = 0.25$

H_1: $p < 0.25$

Calculate the observed sample proportion.

$\hat{p} = \frac{20}{120} = 0.1667$

Under H_0, use $p_0 = 0.25$ in the standard error.

$SE = \sqrt{\frac{p\boxtimes(1-p\boxtimes)}{n}} = \sqrt{\frac{0.25(0.75)}{120}} = 0.0395$

$z = \frac{\hat{p}-p\boxtimes}{SE} = \frac{0.1667-0.25}{0.0395} = -2.1082$

Because H_1 is <, use the lower tail.

p-value = 0.0175

Compare the p-value with $\alpha = 0.1$.

$0.0175 \leq 0.1$, so reject H_0.

There is sufficient evidence at the 10% level to conclude that the proportion of calculators that are defective has decreased from 0.25.

Teacher answers & working

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S2

In a sample of 100, 69 are students choosing the new revision platform. Test at the 1% level whether the proportion of students who choose the new revision platform is different from 0.6. Use a normal approximation for the sample proportion.

Answer:

do not reject H_0

Working:

Let p be the proportion of students who choose the new revision platform.

H_0: $p = 0.6$

H_1: $p \neq 0.6$

Calculate the observed sample proportion.

$\hat{p} = \frac{69}{100} = 0.6900$

Under H_0, use $p_0 = 0.6$ in the standard error.

$SE = \sqrt{\frac{p\boxtimes(1-p\boxtimes)}{n}} = \sqrt{\frac{0.6(0.4)}{100}} = 0.0490$

$z = \frac{\hat{p}-p\boxtimes}{SE} = \frac{0.6900-0.6}{0.0490} = 1.8371$

Because H_1 is two-sided, double the smaller tail probability.

p-value = 0.0662

Compare the p-value with $\alpha = 0.01$.

$0.0662 > 0.01$, so do not reject H_0.

There is insufficient evidence at the 1% level to conclude that the proportion of students who choose the new revision platform is different from 0.6.

Teacher answers & working

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S3

Choose a suitable hypothesis test and carry it out. A sample of 27 gives 6 seeds that germinate. The original claim is that the germination probability is 0.45. Test at the 10% significance level whether the proportion has decreased.

Answer:

reject H_0

Working:

The number of trials is fixed and each trial is success/failure, so an exact binomial test is appropriate.

The probability p is tested directly, and the observed number of successes is compared with $X \sim B(n, p_0)$.

Let p be the proportion of seeds that germinate.

$H_0: p = 0.45$

$H_1: p < 0.45$

Under H_0, $X \sim B(27, 0.45)$. The observed value is $x = 6$.

The alternative hypothesis tells us to use the lower tail.

p-value = $P(X \leq 6) = 0.0125$

Compare the p-value with $\alpha = 0.1$.

$0.0125 \leq 0.1$, so reject H_0.

There is sufficient evidence at the 10% level to conclude that the proportion of seeds that germinate has decreased.

Teacher answers & working

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S4

In a sample of 200, 56 are seeds that germinate. Test at the 1% level whether the proportion of seeds that germinate has increased from 0.25. Use a normal approximation for the sample proportion.

Answer:

do not reject H_0

Working:

Let p be the proportion of seeds that germinate.

H_0: $p = 0.25$

H_1: $p > 0.25$

Calculate the observed sample proportion.

$\hat{p} = \frac{56}{200} = 0.2800$

Under H_0, use $p_0 = 0.25$ in the standard error.

$SE = \sqrt{\frac{p\boxtimes(1-p\boxtimes)}{n}} = \sqrt{\frac{0.25(0.75)}{200}} = 0.0306$

$z = \frac{\hat{p}-p\boxtimes}{SE} = \frac{0.2800-0.25}{0.0306} = 0.9798$

Because H_1 is >, use the upper tail.

p-value = 0.1636

Compare the p-value with $\alpha = 0.01$.

0.1636 > 0.01, so do not reject H_0.

There is insufficient evidence at the 1% level to conclude that the proportion of seeds that germinate has increased from 0.25.

Teacher answers & working

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EA1

In a sample of 80, 44 are defective calculators. Test at the 5% level whether the proportion of calculators that are defective is different from 0.6. Use a normal approximation for the sample proportion.

Answer:

do not reject H_0

Working:

Let p be the proportion of calculators that are defective.

H_0: $p = 0.6$

H_1: $p \neq 0.6$

Calculate the observed sample proportion.

$\hat{p} = \frac{44}{80} = 0.5500$

Under H_0, use $p_0 = 0.6$ in the standard error.

$SE = \sqrt{\frac{p\boxtimes(1-p\boxtimes)}{n}} = \sqrt{\frac{0.6(0.4)}{80}} = 0.0548$

$z = \frac{\hat{p}-p\boxtimes}{SE} = \frac{0.5500-0.6}{0.0548} = -0.9129$

Because H_1 is two-sided, double the smaller tail probability.

p-value = 0.3613

Compare the p-value with $\alpha = 0.05$.

0.3613 > 0.05, so do not reject H_0.

There is insufficient evidence at the 5% level to conclude that the proportion of calculators that are defective is different from 0.6.

Teacher answers & working

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EA2

Choose a suitable hypothesis test and carry it out. A sample of size 81 has mean 74.67 g. The population standard deviation is known to be 20 g. Test at the 10% level whether the population mean packet mass has decreased from 80 g.

Answer:

reject H_0

Working:

The question concerns a sample mean with known standard deviation, so a normal z-test for μ is appropriate.

The sample mean is standardised using its standard error, σ/√n.

Let μ be the population mean packet mass.

H_0: $\mu = 80$

H_1: $\mu < 80$

Under H_0, the sampling distribution of the sample mean has mean 80 and standard error $\frac{\sigma}{\sqrt{n}}$.

standard error = $\frac{20}{\sqrt{81}}$ = 2.2222

Standardise the observed sample mean.

$z = \frac{\bar{x}-\mu\boxtimes}{\sigma/\sqrt{n}} = \frac{74.67-80}{2.2222} =$ -2.4000

Because H_1 is <, use the lower tail.

p-value = 0.0082

Compare the p-value with α = 0.1.

0.0082 ≤ 0.1, so reject H_0.

There is sufficient evidence at the 10% level to conclude that the population mean packet mass has decreased from 80 g.

Teacher answers & working

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EB1

In a sample of 120, 31 are students choosing the new revision platform. Test at the 10% level whether the proportion of students who choose the new revision platform has decreased from 0.3. Use a normal approximation for the sample proportion.

Answer:

do not reject H_0

Working:

Let p be the proportion of students who choose the new revision platform.

H_0: $p = 0.3$

H_1: $p < 0.3$

Calculate the observed sample proportion.

$\hat{p} = \frac{31}{120} = 0.2583$

Under H_0, use $p_0 = 0.3$ in the standard error.

$SE = \sqrt{\frac{p\boxtimes(1-p\boxtimes)}{n}} = \sqrt{\frac{0.3(0.7)}{120}} = 0.0418$

$z = \frac{\hat{p}-p\boxtimes}{SE} = \frac{0.2583-0.3}{0.0418} = -0.9960$

Because H_1 is <, use the lower tail.

p-value = 0.1596

Compare the p-value with $\alpha = 0.1$.

0.1596 > 0.1, so do not reject H_0.

There is insufficient evidence at the 10% level to conclude that the proportion of students who choose the new revision platform has decreased from 0.3.